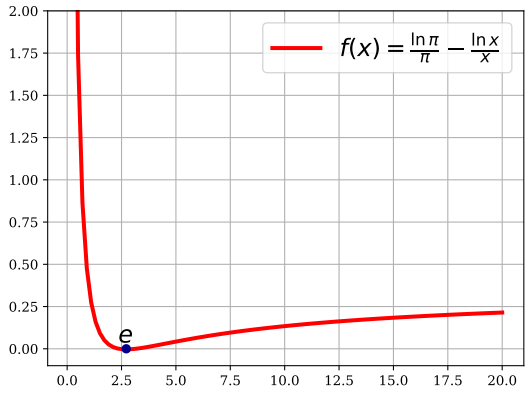
01-EqSolve-exc

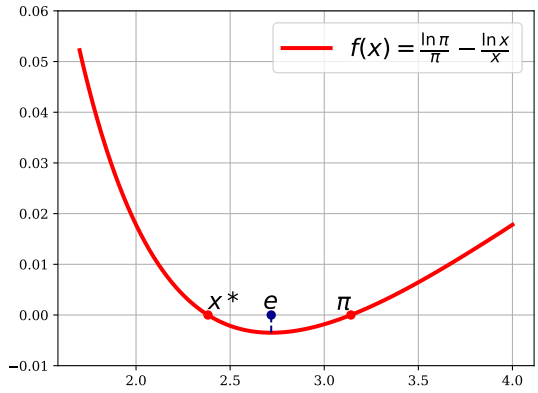
Solving Equations

1. Numme Nisse struggles with the equation $\pi^x = x^\pi$, $x > 0$, which has the apparent root $x = \pi$. Still, he has the tickling feeling that, besides π, there might be other positive roots. As 572738

$$\pi^x = x^\pi \quad \Leftrightarrow \quad x \ln \pi = \pi \ln x \quad \Leftrightarrow \quad f(x) = \frac{\ln \pi}{\pi} - \frac{\ln x}{x} = 0,$$

Nisse plots $f(x)$, then repeats the plot zooming in around the min point $x = e$, *et voilà*:





Nisse's hunch was right, f has a second root, $1 < x^* < e$.

   (a) Help Nisse to construct a Newton-Raphson iterative formula to find that second root x^*.

   (b) Make two iterations starting from $x_0 = 2.37$ to find x^* with 5-6 significant digits.

2. Pär tries a fix-point formula to find the n-th root $\sqrt[n]{a}$ of $a > 0$, by solving the equation

$$x^n = a \quad \text{iterating naïvely} \quad x_{k+1} = \phi(x_k) \quad \text{with} \quad \phi(x) = \frac{a}{x^{n-1}}.$$

Unfortunately, he fails no matter which $a > 0$, $n \geq 2$, and initial guess x_0 he chooses.

   (a) Motivate why Pär's naïve fix point formula does not work.

   (b) He turns serious and constructs iterations $x_{k+1} = \phi(x_k)$, $k = 0, 1, 2, \ldots$, by Newton Raphson's formula. Spell out $\phi(x)$, then use it to make one iteration to find $\sqrt[5]{5}$ starting from $x_0 = 1.37$!

3. It turns out that one can find the inverse a^{-1} of a number $a > 0$ without any division whatsoever by solving the equation $1/x - a = 0$ using Newton-Raphson. 0227a0

   (a) Do so by finding the explicit Newton-Raphson formula $x_{k+1} = \phi(x_k)$, $k \geq 0$, for this equation.

   (b) Make three iterations to approximate $\dfrac{1}{\pi}$ starting with any $0 < x_0 < 0.6$. ($\pi \approx 3.141592653589793$.)

   (*Feather-in-the-hat, but optional*: Can you pinpoint the exact interval $0 < x_0 < b$ for which N-R converges?)

4. The function $\cos(x)$ has a fix point x^* between $\pi/5$ and $\pi/4$. Provided that calculations are performed exactly, motivate in how many iterations can x^* be found up to approximately 15 significant digits e583ea

   (a) by iterating $x = \cos(x)$; (b) by solving $\cos(x) - x = 0$ using Newton-Raphson.

   For (b), write down the formula used and make one Newton-Raphson iteration starting at $x_0 = \pi/4$. e583ea

5. Sara is tries to find as accurately as possible the recursively defined number 404a8e

$$x^* = \sqrt[3]{2\sqrt{3\sqrt[3]{2\sqrt{3\cdots}}}} \quad \rightsquigarrow \quad \{\text{via fix pount iterations}\} \quad \rightsquigarrow \quad \begin{cases} x_0 = 1, \\ x_{n+1} = \phi(x_n) \stackrel{\text{def}}{=} \sqrt[3]{2\sqrt{3x_{n-1}}}. \end{cases}$$

The sequence $\{x_k\}_{k\geq 0}$ is apparently increasing, and, after a couple of iterations, Sara finds that x_k seems to stabilize around $x^* \approx 1.6$. Justify that x^* is an attractor and find analytically in approximately how many iterations will Sara get x^* up to machine precision, ca 15 significant digits.

6. Kalle shows diligently that the recursively defined sequence f50dab

$$x_{k+1} = \sqrt{3\sqrt{5x_k}}, \quad x_1 = 1,$$

is increasing and bounded from above ($x_1 < x_2 < \cdots < x_n < \cdots < 5$), so that $x_k \to x^*$ as $k \to \infty$.

(a) Kalle tries first Fix Point iteration $x_{k+1} = \phi(x_k)$. In approximately how many iterations is x^* located with accuracy $\epsilon \simeq 10^{-12}$? (Hint: $\epsilon_n = |x_n - x^*| = |\phi'(\xi)|\epsilon_{n-1}$, ξ between x^* and x_n.)

(b) As convergence is slow, Kalle applies Newton-Raphson for the root x^* of $f(x) = x - \phi(x)$. How many iterations are needed for the same accuracy? (Hint: $\epsilon_n \simeq C\epsilon_{n-1}^2$, $C = |f''(x^*)/2f'(x^*)|$)

(c) Finally, Kalle finds $x^* = \sqrt{3\sqrt{5\sqrt{3\sqrt{5\cdots}}}}$ analytically. What's the exact value of x^*?

7. Linn köper en soffa för $L = 14\,500$:- på avbetalning med 0% ränta. Upplägningsavgiften är $U = 695$:-. Hon delbetalar i $m = 36$ månader med aviavgift $a = 55$:-/mån, dvs $M = L/m + a$ kr/mån. 9b8024

Effektiva räntan r_e = årsräntan med vilken *lån+avgifter* hade betalats med ränta på nettolånet L.

Teckna den effektiva månadsförräntningen med $f_e = 1 + \sqrt[12]{r_e} > 1$. Linns avbetalninngsplan blir:

Månad	Linn betalar	Netto
0	U	U
1	M	M/f_e
2	M	M/f_e^2
$\vdots$	$\vdots$	$\vdots$
m	M	M/f_e^m
	Summa:	L

$$\begin{aligned} \rightsquigarrow \quad & L = U + M(f_e^{-1} + f_e^{-2} + \cdots + f_e^{-m}) \\ \Leftrightarrow \quad & (L-U)f_e^m + M(f_e^{m-1} + \cdots + f_e + 1) \\ \Leftrightarrow \quad & (L-U)f_e^m = M(f_e^m - 1)/(f_e - 1) \\ \Leftrightarrow \quad & f_e^m(f_e + 1) = b(f_e^m - 1) \\ \Leftrightarrow \quad & \boxed{f_e^{m+1} - (b+1)f_e^m + b = 0}, \; b = \frac{M}{L-U}. \end{aligned}$$

(a) Tag t ex $f_{e_0} = 1.1$ att genom Newton-Raphson bestämma den effektiva månadsförräntningen f_e.

(b) Beräkna den effektiva årsränta $r_e = f_e^{12} - 1$ (som långivaren är enligt lag tvungen at ange).

8. The equation $x = -2\ln x$ has a root $x^* \in (0.5, 1)$. Write down the iterative Newton-Raphson scheme to find the root x^* starting at $x_0 = 0.5$. Further, *either one* of the two alternatives below gives you full points: bebad6

(a) find the root x^* up to 6 significant digits, – *or* –

(b) make an estimate for the *number* of iterations n to obtain x^* with accuracy $\epsilon_n = |x^* - x_n| < 10^{-7}$.

9. Pelle tries to solve the equation $f(x) = x^2 - 6x + 4 = 0$ by substituting $y = x - 3$ and iterating f74275

$$y_{n+1} = \frac{1}{2}(y_n + \frac{5}{y_n}), \quad n = 0, 1, 2, \ldots.$$

(a) Motivate whether Pelle's formula represents the Newton-Raphson method for this equation.

(b) What accuracy $|x^* - x_3|$ can Pelle claim for the root $5 < x^* < 6$ after three iterations of the above scheme if his initial guess is $x_0 = 6$, i.e., $y_0 = 3$?

10. Stina has shown that (e.g., by induction) $x^* = \sqrt{\pi + \sqrt{\pi + \sqrt{\pi + \cdots}}} < 4$, and, ambitious as she is, tries to find it numerically up to at least 10 significant decimals starting at $x_0 = \pi$. 4e5098

(a) Stina tries first Fix Point iteration $x_{k+1} = \sqrt{\pi + x_k} =: \phi(x_k)$. In approximately how many iterations did she succeed? (Hint: $\epsilon_n = |x_n - x^*| \simeq |\phi'(\xi)|\epsilon_{n-1}$ with ξ between x^* and x_n.)

(b) As convergence is slow, she rewrites the equation as $f(x) = x - \sqrt{\pi + x} = 0$, and tries to find the root by Newton-Raphson. Write down the iteration scheme she uses. Approximately how many iterations did it take this time? (Hint: $\epsilon_n \simeq C\epsilon_{n-1}^2$, with constant $C \simeq \dfrac{|f''(x^*)|}{2|f'(x^*)|}$ for large n.)

(c) Finally, in a glimpse of revelation, Stina squares the equation, and finds trivially x^* (sigh…).

11. Finding the root x^* of $f(x) = 0$ by the Fix Point method requires rewriting the equation as $x = \phi(x)$ and iterate starting at x_0 near x^*. Convergence requires $|\phi'(x)| < 1$. A plausible rewrite with $|\phi'| \ll 1$ will be ff1530

$$f(x) = 0 \;\Leftrightarrow\; x = x + cf(x) =: \phi(x). \qquad (\star)$$

(a) Having the initial guess x_0 reasonably close to x^*, set c in such a way that $\phi'(x_0) = 0$.

(b) Iterate $(\star)$ suitably to find the root x^* of $x^5 = 10x + 1$ near $x_0 = 2$ until $|x_{k+1} - x_k| < 0.01$.

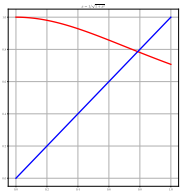
12. Given are two forms of the equation $x^5 = 10x + 1$ with apparent root $1 < x^* < 2$: ff1531

$$\text{(a) } x = \frac{x^5 - 1}{10} \qquad \text{and} \qquad \text{(b) } x = (10x + 1)^{1/5}.$$

Motivate which of them is suitable to find x^* by fix point iterations $x_{k+1} = \phi(x_k)$ if one starts at $x_0 = 1.5$ and why? Make an estimate how many iterations would be needed to reach accuracy $\epsilon_n = |x_n - x^*| < 10^{-6}$?

13. Putte tries to solve by the Newton-Raphson's method the equation $e^{-x^2} = 0.1x$, which has apparently a single positive root. Unfortunately, starting at $x_0 = 0$, he obtains cyclically 0, 10, 0, 10, 0, …. Explain in detail what goes wrong with his setting and suggest a possible remedy. 050424

14. Joe figures out that the equation $x = 1/\sqrt{1 + x^2}$ has a single root x^*, $0 < x^* < 1$. He wants to find it quickly and decides to use Newton Raphson's method. After a bit of algebra, Joe comes up with the iteration scheme a9736b



$$x_{n+1} = \frac{1 + 2x_n^2}{x + (1 + x_n^2)^{3/2}},\ n = 0, 1, 2, 3, \ldots.$$

(a) Check whether Joe's formula really represents the Newton-Raphson scheme to solve this equation.

(b) Find the root of the equation with about 6 significant digits, when $|x_{n+1} - x_n| < 10^{-6}$.

Roots of $\mathbb{C}$-valued functions

1. (Optional) Use Newton-Raphson with start from different z_0 to find all three roots of f8f587

$$f(z) = z^3 + (-5 + i)z^2 + (13 - 2i)z - (11 + 3i).$$

You can use the Matlab script below to play with (e.g, in a browser; google: `Octave online`):

```
NewtR = @(z,f,fp) z - f(z)/fp(z);     % Newton-Raphson
a = -5+i; b = 13-2i; c = -(11+3i);    % The coeff
f  = @(z) z.^3 + a*z.^2 + b*z + c;    % f(z)
fp = @(z) ...                         % <-- Fill in f'(z)
z = ...                               % <-- Fill in initial pt z0
for j = 1:10
  z  = NewtR(z,f,fp);
  printf("%3d & %10.6f%+10.6fi \n", j,real(z),imag(z))
end
```

Non-linear Systems

1. Rewrite the non-linear system 49ec9f

$$\begin{cases} 3x_1 - x_2 + \ln(x_1^2 + x_2^2) = 23 \\ x_1^2 - x_1x_2 + 3x_2^2 = 41 \end{cases}$$

in the form $\mathbf{f}(\mathbf{x}) = \mathbf{0}$. Then devise a Newton-Raphson scheme for finding its roots,

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - J^{-1}(\mathbf{x}^{(k)})\mathbf{f}(\mathbf{x}^{(k)}), \quad k = 0, 1, 2, \ldots,$$

and compute the first iteration $\mathbf{x}^{(1)}$ to find the root $\mathbf{x}^*$ in the vicinity of $\mathbf{x}^{(0)} = (7, 1)$.

2. Construct the Newton-Raphson scheme to approximate the solution $\mathbf{x}^* > \mathbf{0}$ of the non-linear system: 77a221

$$\begin{cases} x_1x_2/3 + x_2^2/2 - 6x_1 + 24x_2 &= 387, \\ 9x_1^2/2 + 20x_1x_2/3 + 387x_1 - 129x_2 &= -129. \end{cases}$$

Compute the first approximation $\mathbf{x}^{(1)}$ of $\mathbf{x}^*$ starting at the initial guess $\mathbf{x}^{(0)} = (3, 4)$.

3. Write down the vector field $\mathbf{f}(\mathbf{x})$ and Jacobian matrix $J(\mathbf{x})$ for the Newton iteration scheme 78a542

$$\begin{cases} \mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \mathbf{h}^{(k)} \\ J(\mathbf{x}^{(k)})\mathbf{h}^{(k)} = -\mathbf{f}(\mathbf{x}^{(k)}) \end{cases} \quad \text{to solve the system} \quad \begin{cases} 4.7\sin(2x_1) + 3.1e^{x_2} = 1.5 \\ 3.6\cos(3x_1) + 1.2\sin(x_2) = 2.4 \end{cases}$$

given an initial point $\mathbf{x}^{(0)}$ near a root. Make one iteration starting at $\mathbf{x}^{(0)} = [4.4, -3.0]^T$.

Parallelized

1. Putte's iterations when solving the equation $f(x) = 0.01x - \cos(0.005x) = 0$ steady down giving a function value of $f(x_n) = -1.2 \cdot 10^{-15}$ and better no further. What accuracy can Putte claim for x_n as an approximation for the root, given that the rounding errors in computing f are $\delta \leq 5 \cdot 10^{-15}$? ffeb71

2. Putte tries to solve numerically the equation $f(x) = 0$ with a known root x^*, where $f \in C^2$ near the root. Pute starts at $x_0 = x^* + 0.1$ and, using the Newton-Raphson method, obtains errors 816043

$$|x_1 - x^*| \simeq 1.3 \cdot 10^{-3} \quad \text{and} \quad |x_2 - x^*| \simeq 1.73 \cdot 10^{-7}.$$

Provided rounding errors are negligible, what error $|x_3 - x^*|$ should he epxect at iteration three?

3. Motivate that the equation 10f27b

$$x = \arcsin(x - 0.2) \qquad (\star)$$

   (a) has a root $x^* \in [-0.8, 0.8]$? Is the root unique?

   (b) Can x^* be found iterarively using that it is a fix point for the rhs of $(\star)$?

   (c) If not, whiy? Rewrite $(*)$, so that it can be solved by the fix point iteration? Specify explicitly the function φ for which $x^* = \lim_{n\to\infty} \varphi(x_n)$.

4. Consider the equation $f(x) = x^2 - 3 = 0$. Find a function φ, such that the iteration $x_{i+1} = \varphi(x_i)$, $i = 0, 1, 2 \ldots$, realizes the Newton-Raphson method to find the positive root $x^* = \sqrt{3}$ of f. Make a sketch of $f(x)$ and motivate graphically why convergence is guaranteed from any initial $x_0 > 0$? 522b38

   (a) Make a 3-4 iterations by hand starting at $x_0 = 3$.

   (b) Make a 5-6 iterations with $x_0 = 0.1$. Why is x_1 farther away from the root $x^* = \sqrt{3}$ than x_0?

5. Consider the non-linear system (the intersection pt of two hyperbolas): $\begin{cases} x_1^2 - x_2^2 = 11 \\ x_1x_2 = 30, \ x_1 > 0, \end{cases}$ 17d63d

(a) Express x_1 from the first equation, and x_2 from the second to reformulate it as a fixed point problem

$$\mathbf{x}^{[j+1]} = \boldsymbol{\phi}(\mathbf{x}^{[j]}),\ j \geq 0. \qquad (\star)$$

(b) Find $J_{\boldsymbol{\phi}}(\mathbf{x}^*) = \dfrac{d\boldsymbol{\phi}}{d\mathbf{x}}(\mathbf{x}^*)$ at the solution $\mathbf{x}^* = (6, 5)$ and its eigenvalues by solving $|J - \lambda I| = 0$.

(c) Compute $|J_{\boldsymbol{\phi}}(\mathbf{x}^*)| = \max\limits_i |\lambda_i|$ (λ does not have to be real!) and rule whether $\mathbf{x}^*$ is an attractor and thus can be found by $(\star)$ starting at $\mathbf{x}^{[0]}$ sufficiently near the root $\mathbf{x}^*$.

(d) Play withe Matlab code ….